

# Daniel W Liu

## Computer Science and Finance Student

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### EDUCATION

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| <b>University of Waterloo</b>   <i>Double Major in Computer Science and Finance</i><br><i>Bachelor of Computing and Financial Management (Co-op)</i> — <b>GPA: 3.94/4.0</b> | Sept 2025 – April 2029<br>Waterloo, ON |
| <b>International Baccalaureate Diploma Programme</b><br><i>IB Diploma</i> — <b>45/45 Perfect Score</b> — <i>Top 0.5% Internationally</i>                                    | Sept 2021 – May 2025<br>Thornhill, ON  |

### TECHNICAL SKILLS

**Languages:** Python, C, C++, JavaScript, TypeScript, GLSL, SQL, HTML, CSS  
**Frameworks:** React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS  
**Cloud and Databases:** AWS, Google Cloud, PostgreSQL, MongoDB, Redis, Supabase, Drizzle  
**Data and Libraries:** OpenGL, pandas, NumPy, OpenCV, Matplotlib, yFinance, PyQt5, Three.js  
**DevOps and Tools:** Docker, Git, Linux, CI/CD, Jest, Jenkins, Vercel, GitHub, Unity, Blender

### EXPERIENCE

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| <b>Software Developer Intern - Builds, Infrastructure and Tooling</b><br><i>PlayStation (Sony Interactive Entertainment)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | May 2026 – Aug 2026<br>Waterloo, ON      |
| <b>Software Engineering Intern</b><br><i>Wedge (YC S25)</i> <ul style="list-style-type: none"><li>Shipped a full-stack AI voice platform using Next.js, Supabase, and Vapi, meeting weekly with <b>3</b> hospital clients to auto-navigate insurance claim calls and extract <b>45+ structured claim data fields</b> across <b>50+ clinical sites</b>.</li><li>Reduced <b>server polling overhead by 67%</b> and resolved race conditions through atomic Postgres operations, while building an Interactive Voice Response engine that auto-maps call flows into <b>8-node AI decision trees</b>.</li></ul>                | Jan 2026 – Apr 2026<br>San Francisco, CA |
| <b>Software Engineering Intern</b><br><i>HOSA Canada</i> <ul style="list-style-type: none"><li>Engineered a scalable full-stack educational resource sharing platform using Next.js, AWS, and MongoDB, serving <b>9,000+ Canadian students</b> competing across <b>50+ unique associations</b> spanning North America and Asia.</li><li>Deployed a production MERN workshop sign-up system processing <b>19,943 registrations</b> across <b>347 workshop sessions</b>, integrating Twilio email notifications, Redis queues, PM2 cron workers, and <b>457 Jest unit tests</b>.</li></ul>                                   | Sept 2025 – Dec 2025<br>Toronto, ON      |
| <b>Machine Learning Research Intern</b><br><i>Pearl Lab Holland Bloorview Kids Rehabilitation Hospital</i> <ul style="list-style-type: none"><li>Developed real-time AI object detection systems using OpenCV and pupil-apriltags for <i>Bootle Blast</i>, a gamified rehabilitation platform clinically tested at <b>3+ Canadian hospitals</b> for children's motor skill rehabilitation.</li><li>Benchmarked ML object detection models using NumPy and pandas against YOLOv8 on a dataset of <b>250+ unique environments</b>, improving <b>mean average precision (mAP) by 46.8%</b> in real-world scenarios.</li></ul> | Jul 2025 – Aug 2025<br>Toronto, ON       |

### PROJECTS

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| <b>rainbolt.ai</b>   <i>Next.js, Three.js, FastAPI, OpenAI CLIP, Pinecone, LangChain, Google Cloud</i> <a href="#">Devpost</a> <a href="#">GitHub</a> <ul style="list-style-type: none"><li>Awarded <b>Best UI Hack</b> at UofT Hack the Valley X as <b>frontend lead</b>, building an AI geolocation teaching platform with Next.js, Three.js, Zustand, and Gemini for interactive 3D visualizations and geographic reasoning.</li><li>Architected a RAG pipeline with a dataset of <b>955,199+ images</b>, integrating OpenAI CLIP, Pinecone, LangChain, and Mapillary API with a FastAPI backend to analyze street-level imagery and generate visual location reasoning.</li></ul> |
| <b>Multithreaded Voxel Engine</b>   <i>C++20, OpenGL 4.1, GLFW, GLM, Dear ImGui, FastNoiseLite, CMake</i> <a href="#">GitHub</a> <ul style="list-style-type: none"><li>Built a multithreaded C++20/OpenGL voxel engine with procedural terrain, shadows, and water, using a greedy meshing algorithm to merge identical block faces and cut GPU triangles <b>27x</b> per chunk, sustaining <b>145 FPS</b>.</li><li>Engineered a chunk-streaming pipeline with <b>9 worker threads (940 chunks/sec)</b>, infinite world streaming with bounded memory, raycast-based block place/break, and <b>RLE-compressed binary saves (144x compression)</b>.</li></ul>                           |
| <b>VibeTrade</b>   <i>Next.js, FastAPI, Supabase, LangGraph, Alpaca, Finnhub, ElevenLabs</i> <a href="#">Devpost</a> <a href="#">GitHub</a> <ul style="list-style-type: none"><li>Built an agentic AI trading dashboard that analyzes market data, news, and portfolio context to generate real-time actionable insights and explainable trading rationales, integrating <b>5 financial APIs</b> over live WebSocket streams.</li><li>Implemented a customizable 3D AI voice agent with low-latency voice synthesis, background workers for market monitoring, and a FastAPI + LangGraph workflow enabling voice-triggered trade execution and reasoning.</li></ul>                   |